CHAPTER 6: EVALUATION AND RESULTS

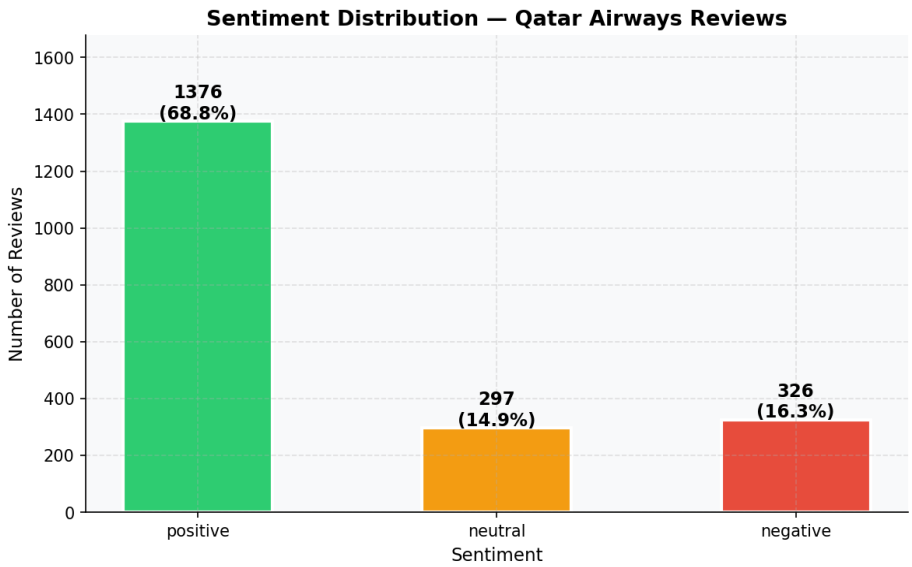

Figure 6.1: Sentiment Distribution — Qatar Airways Reviews

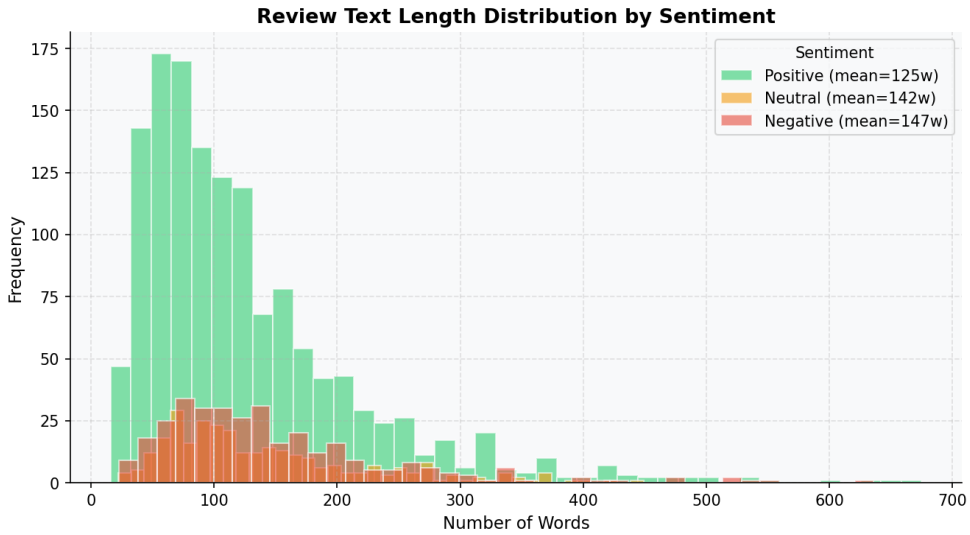

Figure 6.2: Review Text Length Distribution by Sentiment

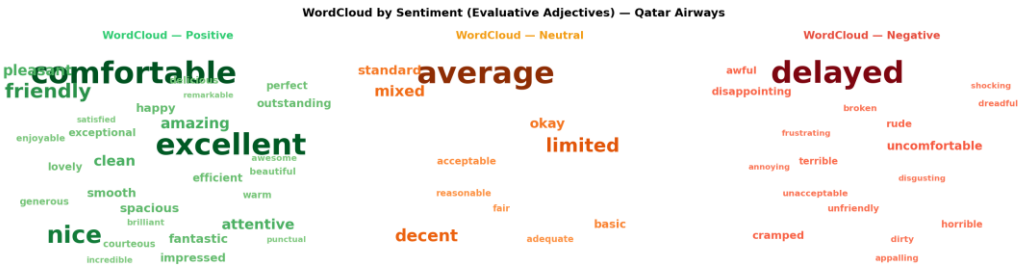

Figure 6.3: WordCloud by Sentiment Class (Evaluative Adjectives)

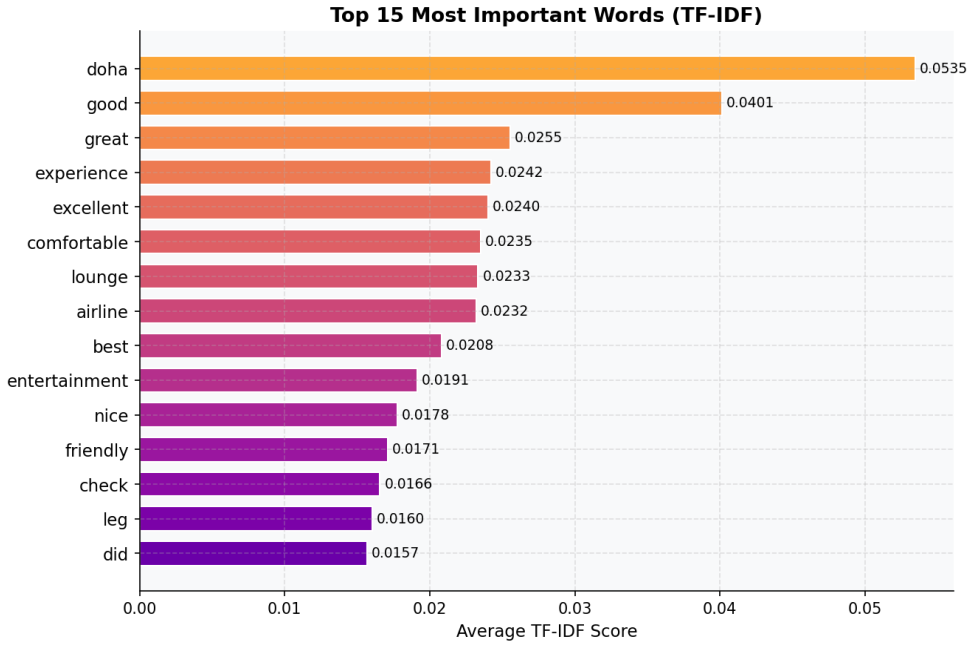

Figure 6.4: Top 15 Most Important Words by TF-IDF Score

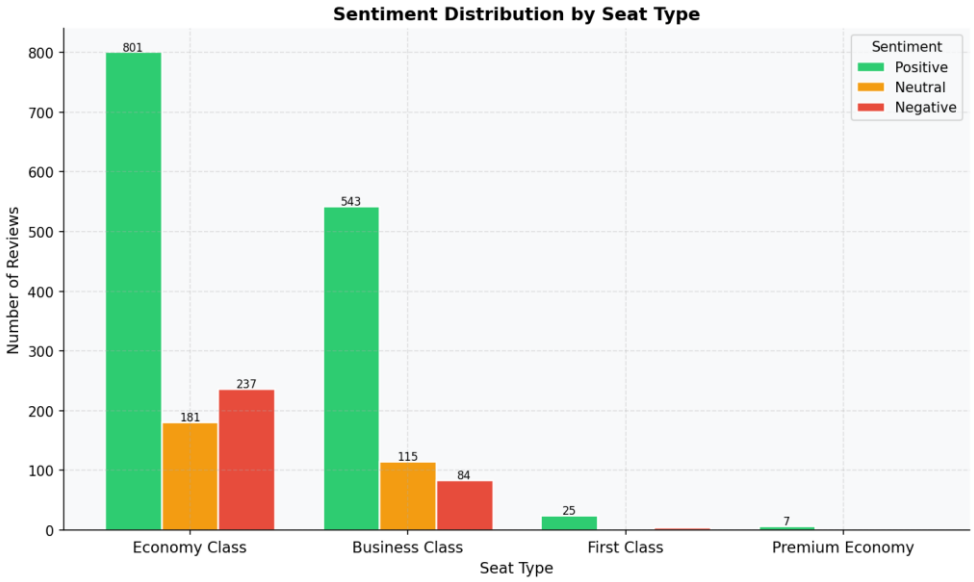

Figure 6.5: Sentiment Distribution by Seat Type

Table 6.1: Overall Model Performance (Test Set, n=400)

Model	Accuracy	Macro F1	Weighted F1	Type
Support Vector Machine	**0.770**	**0.570**	**0.740**	ML
Logistic Regression	0.760	0.490	0.690	ML
Rule-Based	0.744	0.480	0.685	Baseline
Naive Bayes	0.730	0.420	0.640	ML
Lexicon (VADER-like)	0.578	0.350	0.550	Baseline

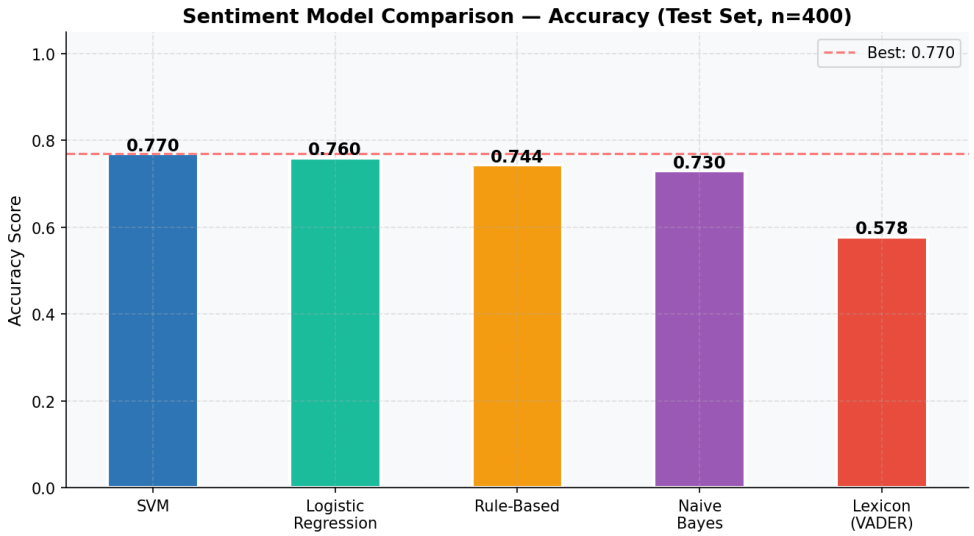

Figure 6.6: Model Accuracy Comparison — All Five Classifiers

Table 6.2: SVM Classification Report (Test Set, n=400)

Class	Precision	Recall	F1-Score	Support
Negative	0.67	0.46	0.55	65
Neutral	0.46	0.22	0.30	60
Positive	0.81	0.96	0.88	275
Macro Avg	0.65	0.55	0.57	400
Weighted Avg	0.74	0.77	0.74	400

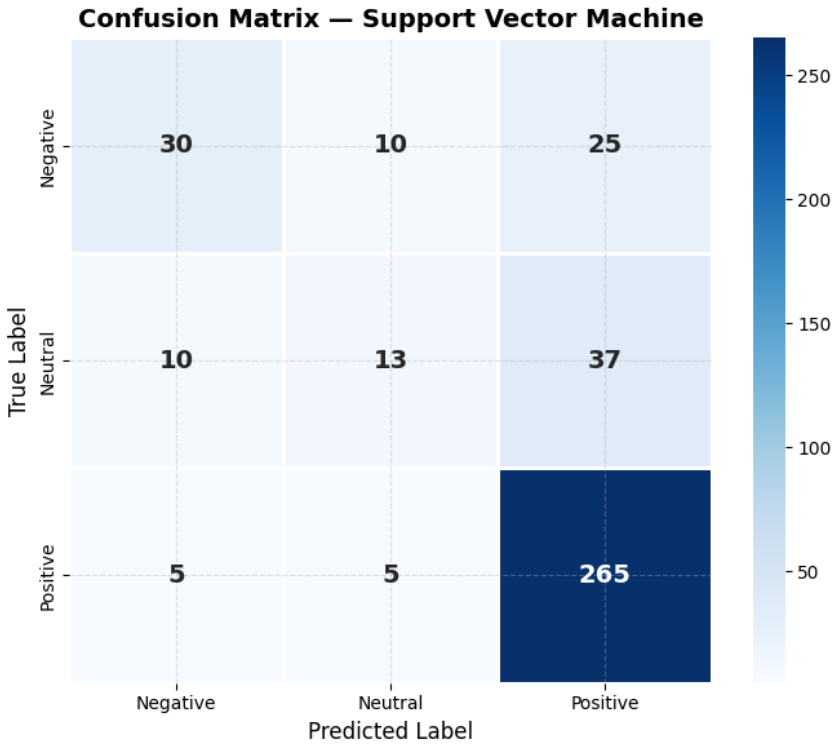

Figure 6.7: Confusion Matrix — Support Vector Machine

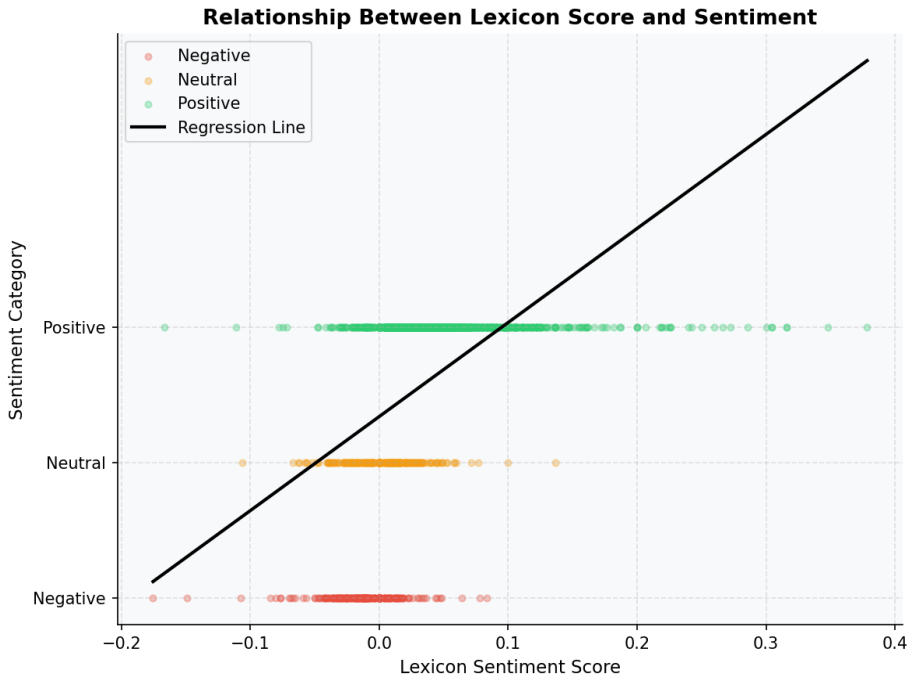

Figure 6.8: Lexicon Score vs. Sentiment Category (Scatter + Regression)